



Project Proposal

**Subject: Mobile Application
Development (MAD)**

Section: SE 5-1

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Project Title:

Blood Donor Finder & Emergency Blood Assistance App

1. Introduction:

Our project is a mobile application called **Blood Donor Finder & Emergency Blood Assistance App**. The main purpose of this application is to help people find blood donors quickly when they or their family members need blood.

In emergency situations, finding a suitable blood donor can be very difficult. People usually contact relatives, friends, hospitals, WhatsApp groups, or post on social media. This process can take a lot of time, especially when a specific blood group is required.

Our application will provide a simple platform where people can search for available blood donors according to their blood group and location. Donors can register themselves in the application and receive blood requests when someone needs their blood group.

2. Problem Statement:

Finding blood during an emergency is a serious problem. Sometimes patients need blood immediately, but their family members do not know where to find a suitable donor.

There is no simple and centralized system where a person can search for available donors according to blood group, location, and availability.

Because of this, valuable time can be wasted during emergencies. Our project aims to solve this problem by providing a mobile application

that connects blood seekers with suitable blood donors.

3. Proposed Solution:

We propose a mobile application that will connect **blood donors and blood seekers** on one platform.

A person who needs blood will be able to open the application, select the required blood group and location, and search for available donors.

A donor will be able to create a profile, enter their blood group and location, and turn their availability on or off.

When a blood request is created, suitable donors can receive a notification and decide whether they can help or not.

The application will also provide an **Emergency Request** option so that urgent blood requirements can be highlighted.

4. Main Features:

The main features of our application will be:

- User registration and login
- Donor registration
- Blood group selection
- Donor profile
- Search donors by blood group
- Search donors by location
- Donor availability status
- Create blood requests
- Emergency blood requests
- Notifications to suitable donors
- Accept or reject blood requests

- Request status tracking
 - Request history
 - User profile management
 - Admin management system
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5. Users of the Application:

There will be three main types of users:

Blood Donor:

A donor can register in the application, enter their blood group and location, set their availability, and receive blood requests.

Blood Seeker:

A blood seeker can search for suitable donors and create blood requests when blood is required.

Administrator:

The administrator will manage users, verify donors, monitor blood requests, and remove inappropriate or fake accounts.

6. Technologies

We will use the following technologies to develop the project:

- **Flutter** – for developing the mobile application
 - **Dart** – programming language
 - **Firebase Authentication** – for login and registration
 - **Cloud Firestore** – for storing application data
 - **Firebase Cloud Messaging** – for notifications
 - **Android Studio / VS Code** – for development
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7. How the Application Will Work:

The basic working of the application will be:

1. The user creates an account.
 2. The user logs into the application.
 3. A person can register as a blood donor or use the application as a blood seeker.
 4. Donors enter their blood group, location, and availability.
 5. A blood seeker searches for a required blood group.
 6. The application shows suitable available donors.
 7. The blood seeker can create a blood request.
 8. Suitable donors receive a notification.
 9. A donor can accept or reject the request.
 10. The blood seeker can see the request status.
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8. Objectives:

The main objectives of our project are:

- To make finding blood donors easier.
 - To reduce the time required to find a suitable donor.
 - To connect blood donors and blood seekers through one platform.
 - To provide an emergency blood-request facility.
 - To allow donors to control their availability.
 - To provide notifications for blood requests.
 - To maintain blood donor and request information.
 - To develop a simple and user-friendly mobile application.
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9. Expected Benefits:

This application can be helpful in situations where

blood is needed urgently.

The main benefits are:

- People can find suitable donors more easily.
- Donors can know when someone needs their blood group.
- Emergency requests can be highlighted.
- Users can search donors according to location.
- Blood requests can be tracked.
- All important information can be managed through one application.

The application can save time and make communication between blood donors and blood seekers easier.

10. Future Improvements:

In the future, we can add more features to make the application more useful, such as:

- GPS-based nearby donor search
 - Google Maps integration
 - Hospital accounts
 - Hospital verification
 - Blood donation history
 - Urdu language support
 - Donor eligibility reminders
 - Blood bank integration
 - Automatic donor matching
 - Advanced emergency notifications
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11. Conclusion:

The Blood Donor Finder & Emergency Blood Assistance

App is a real-world mobile application project designed to help people find blood donors more easily.

The main idea is to bring blood donors and blood seekers onto one platform. Users will be able to search for donors, create blood requests, receive notifications, and manage their requests.

We will develop the application using **Flutter and Dart**, with **Firestore** for authentication, database management, and notifications.

This project will help us apply our knowledge of mobile application development while working on a real-life problem that can benefit people during emergency situations.

